MIL-F-14902A(PA)
23 November 1973
SUPERSEDING
MIL-F-14902(MU)
1 August 1968

MILITARY SPECIFICATION

FIN, ASSEMBLY, 60MM M2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of fin assembly for 60MM mortar ammunition designated as Fin, Assembly, 60MM, M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 -Ammunition and Special Weapons, General Specification for

STANDARDS

FEDERAL

Federal Test Method

Standard No. 151 - Metals, Test Methods

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 -Quality Assurance Terms and Definitions
MIL-STD-1168 -Lot Numbering of Ammunition

FSC: 1310

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MIL-STD-1169 -Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 -Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PRODUCT AND PACKING DRAWINGS

9207612 -Fin Assembly, 60MM, M2

INSPECTION EQUIPMENT DRAWINGS

9221600 -Location and Alignment Gage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The fin assembly shall comply with all requirements specified on Drawing (dwg.) 9207612 and with all requirements specified in applicable specifications.

3.3 Strength of assembly..-The load required to pull each fin from the cartridge container shall be 1,700 pounds, minimum, when tested in accordance with 4.4.1.

3.4 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.5 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible

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for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Initial production inspection

4.2.1 Submission.-Prior to initiation of regular production, the contractor shall submit an initial production sample to a Government approved laboratory designated by the contracting officer (see 6.3) for evaluation in accordance with the provisions 4.2.2.

The initial production sample shall consist of fifteen painted assemblies, ten unpainted assemblies, and ten complete sets of parts (i.e, ten each of every component) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract.

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In addition the contractor shall furnish two one foot lengths of the bar stock which is being used to fabricate the cartridge container. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2. Inspections to be performed.-Assemblies and components will be subjected to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings. The five fin assemblies (those with no flash holes and threads) shall be subjected to tests for the physical properties referenced on dwgs. 9207544 and 9207545.

4.2.3 Rejection.-If any assembly or component fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer in one unchanged

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process, using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specification, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235, may be used if approved by the Government. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Fin (see dwg. 9207545 covering a detail of dwg. 9207612).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
101.	Material thickness	Gage (CD)
Minor:		
201.	Radius between web and blades, minimum (min.)	Gage (CD)
202.	Maximum blade radius forward end	Gage (CD)
203.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.2 Container cartridge (see dwg. 9207544 covering a detail of dwg. 9207612).

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Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
101.	Depth of ignition cartridge cavity cavity, min	Gage (CD)
102.	Pitch diameter of external thread, min	Gage (CD)
103.	Major diameter of external thread, min	Gage (CD)
104.	Perpendicularity of shoulder to external thread	Gage (CD)
105.	Pitch diameter of internal thread, max	Gage (CD)
106.	Minor diameter of internal thread, max	Gage (CD)
107.	Concentricity of internal thread to external thread	Gage (CD)
108.	Concentricity of outside diameter to external thread	Gage (CD)
109.	Depth of perfect internal threads, min	Gage (CD)
110.	Diameter of ignition cartridge cavity, min	Gage (CD)
111.	Diameter of external thread undercut, min	Gage (CD)
112.	Width of external thread undercut, max	Gage (CD)
113.	Diameter of rear counterbore, min	Gage (CD)
114.	Depth of counterbore	Gage (CD)
Minor:		
201.	Length from end of external thread to shoulder	Gage (CD)
202.	Location of first row of holes	Gage (CD)
203.	Diameter of cartridge container	Gage (CD)
204.	Location of second row of holes	Gage (CD)
205.	Depth of primer cavity, max	Gage (CD)
206.	External thread undercut radius	Gage (CD)
207.	Diameter of flash holes, min	Gage (CD)
208.	Pitch diameter of internal thread, min	Gage (CD)
209.	Pitch diameter of external thread, max	Gage (CD)
210.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.3 Fin, assembly, prior to painting (see dwg. 9207612).

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Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	100 percent inspection	
101.	Fin blade(s) bent, distorted or mislocated (functional)	Gage 9221600
Major:		
102.	Length of assembly, max	Gage (CD)
103.	Diameter of assembly	Gage 9221600
104.	Concentricity of fin projections with external thread	Gage (CD)
105.	Projection of blades, min	Gage (CD)
106.	Location of rear fin radius to rear face of cartridge container	Gage (CD)
107.	Braze missing or not continuous	Visual
Minor:		
201.	Flash hole(s) missing or obstructed	Visual
202.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.4 Fin assembly, after painting (see dwg. 9207612).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
101.	Threads damaged or painted	Visual
Major:		
102.	Fin blade(s) bent, distorted or mislocated (functional)	Gage 9221600
Minor:		
201.	Protective coating with aggregate of bare spots in excess of 1/4 square inch	Visual
202.	Marking incorrect, incomplete or illegible	Visual
203.	Evidence of poor workmanship (see 3.4)	Visual

4.3.3 Testing

4.3.3.1 Pull test (see 3.3)-Major defect

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4.3.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, a sample of 80 randomly selected assemblies from each lot shall be subjected to this test. If one or more samples fail to comply with the applicable requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.1.2 After three consecutive lots have met the criteria of 4.3.3.1.1, fifty sample fins shall be selected from each lot for this test. If one or more samples fail to comply with the applicable requirements the lot shall be rejected. Testing shall be performed as specified in 4.4.1.

4.3.3.2 Fin blade and cartridge container hardness.-(See dwg. 9207612)-Major defect.-Twenty sample fin assemblies shall be selected from each lot for this test. If one or more samples fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.3 (see 6.2).

4.4 Test methods and procedures

4.4.1 Pull test.-A pair of opposing fins on each assembly shall be assembled to a suitable fixture and the required pull load applied in a direction at right angles to the axis of the fin assembly. Equipment for this test shall be as specified in 4.3.4.

4.4.2 Hardness.-Hardness readings shall be taken at the heat affected areas. A minimum of 3 equi-distance readings shall be taken along one fin, and a minimum of 3 readings taken on the cartridge container. The rockwell hardness tests shall be performed in accordance with Federal Test Method Standard No. 151, Method 243.

5. PREPARATION FOR DELIVERY

5.1 Packing

Level C - Fin assemblies shall be packed in accordance with the requirements of MIL-STD-1169.

5.2 Marking.-Marking shall be in accordance with MIL-STD-1169.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designated by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designns are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The

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contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number, paragraph number and defect number if such number has been assigned shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Submission of initial production samples.-Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA.

Custodian:
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Preparing Activity:
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